



TRAINING PROGRAMME

Introduction to Networks (CCNA 1)

Build, address and secure a small network with Cisco devices

39 HOURS

FOR Students, technicians and IT staff starting the CCNA path

PREPARES FOR CCNA (200-301 v1.1), course 1 of 3



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Overview

This programme is the first of three courses that prepare the Cisco CCNA 200-301 exam, aligned with Cisco Networking Academy Introduction to Networks v7. Participants learn how networks are built and how data moves through them, configure Cisco switches and routers, design IPv4 addressing plans with subnetting and VLSM, deploy IPv6, and secure devices with strong passwords and SSH. Every chapter ends with a lab in Cisco Packet Tracer, and the programme closes with a final project in which each participant designs, builds and documents a small business network. Switching, Routing & Wireless Essentials (CCNA 2) follows.

At a glance

#	PHASE	FOCUS	HOURS
01	Network Fundamentals	Network types, reference models, first device configuration	6 h
02	From Links to Packets	Media, Ethernet and switching, IP packets, ARP and Neighbor Discovery	6 h
03	Routers & IP Addressing	Router configuration, IPv4, subnetting and VLSM, IPv6	9 h
04	Transport, Services & Security	ICMP, TCP and UDP, DNS, DHCP, HTTP, securing a small network	6 h
05	Final Project	A small business network, documented and presented	6 h
	Buffer	Catch-up, revision, extra lab time	6 h
	Total		39 h

Key facts

AUDIENCE	Students, technicians and IT staff starting the CCNA path
PREREQUISITES	None. Basic computer use; comfort with binary numbers helps and is reviewed in the course
DURATION	39 h: 33 h of chapters + 6 h buffer
FORMAT	Short lectures followed by guided labs on simulated Cisco devices
LAB ENVIRONMENT	Runs locally: one laptop per participant with Cisco Packet Tracer (free with each participant's Cisco Networking Academy account); physical Cisco switches and routers for selected labs when the organiser provides a rack; no cloud account or cost
ASSESSMENT	A lab validated at the end of each phase and a final project presented to the group; certificate of completion for participants who complete both
OUTCOME	Participants can build a small switched and routed network, plan IPv4 and IPv6 addressing with subnetting and VLSM, secure Cisco devices and troubleshoot basic connectivity

PHASE 01

6 H

Network Fundamentals

Participants learn what a network is made of and how the OSI and TCP/IP models describe it, then take their first steps in Cisco IOS on switches and end devices. Validated lab: a switch with a hostname, passwords, a management address and a saved configuration, reachable from a PC.

#	CHAPTER	CONTENT	FORMAT	HOURS
1.1	Networks & Models	Components, topologies, network types, OSI and TCP/IP models, standards	Lecture + Lab	3 h
1.2	Basic Device Configuration	Cisco IOS, switch and end-device configuration	Lecture + Lab	3 h

PHASE 02

6 H

From Links to Packets

Participants follow data from the cable to the packet: media and Ethernet frames, how a switch learns MAC addresses, then how IPv4 and IPv6 packets are routed and how addresses are resolved. Validated lab: a captured exchange that shows a MAC table filling, an ARP request and an IPv6 neighbor solicitation, on hosts addressed in advance by the trainer, observed in Packet Tracer simulation mode.

#	CHAPTER	CONTENT	FORMAT	HOURS
2.1	Physical & Data Link Layers	Media, Ethernet frames, MAC addresses, switching, MAC address tables	Lecture + Lab	3 h
2.2	Network Layer & Address Resolution	IPv4 and IPv6 packets, routing, ARP and IPv6 Neighbor Discovery	Lecture + Lab	3 h

PHASE 03

9 H

Routers & IP Addressing

Participants configure a router as the default gateway, then learn IP addressing in depth: IPv4 structure and binary, subnetting and VLSM design, IPv6 address types, SLAAC and IPv6 subnetting. Validated lab: a VLSM addressing plan for a site with several LANs behind one router, implemented in dual stack and verified with ping and traceroute.

#	CHAPTER	CONTENT	FORMAT	HOURS
3.1	Basic Router Configuration	Initial router settings, default gateway	Lecture + Lab	1.5 h
3.2	IPv4 Addressing	Address structure, binary, unicast, broadcast and multicast, public and private addresses	Lecture + Lab	1.5 h
3.3	Subnetting & VLSM	Subnetting IPv4 networks, VLSM design	Lecture + Lab	3 h
3.4	IPv6 Addressing	Address types, SLAAC, subnetting IPv6	Lecture + Lab	3 h

PHASE 04

6 H

Transport, Services & Security

Participants see how applications use the network through ICMP, TCP, UDP, DNS, DHCP and HTTP, then secure the devices of a small network against common threats with strong passwords and SSH. Validated lab: a small network where hosts get addresses from DHCP and reach a web server by name, and every device is managed only over SSH.

#	CHAPTER	CONTENT	FORMAT	HOURS
4.1	Transport & Application Layers	ICMP, TCP and UDP, DNS, DHCP, HTTP	Lecture + Lab	3 h
4.2	Security Basics & Small Networks	Threats, passwords, SSH, securing devices, building a small network	Lecture + Lab	3 h

PHASE 05

6 H

Final Project

Each participant designs and builds the network of a small business from a written brief, reusing the phase labs, then documents it and troubleshoots faults injected by the trainer.

#	CHAPTER	CONTENT	FORMAT	HOURS
5.1	Brief & Build	Read the brief, write the VLSM addressing plan, build the topology in dual stack IPv4 and IPv6	Project	3 h
5.2	Harden & Document	Device hardening and SSH access, addressing table, topology diagram, saved configurations	Project	1.5 h
5.3	Troubleshoot & Review	Find and fix faults injected by the trainer; questions and feedback	Project	1.5 h

FINAL PROJECT

Small Business Network

The network of a small business with a VLSM addressing plan, IPv4 and IPv6 connectivity, hardened Cisco devices managed over SSH, and documentation that lets another technician troubleshoot it.

STACK Cisco Packet Tracer, Cisco IOS switches and routers, IPv4, IPv6, SSH

DELIVERABLE Packet Tracer file, addressing table, topology diagram, device configurations